Cross-Modal Contrastive Learning for Remote Sensing Image Classification

Zhixi Feng, *Member, IEEE*, Liangliang Song, Shuyuan Yang, *Senior Member, IEEE*, Xinyu Zhang, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— Recently, multimodal remote sensing image (MRSI) classification has attracted increasing attention from researchers. However, the classification of MRSI with limited labeled instances is still a challenging task. In this article, a novel self-supervised cross-modal contrastive learning (CMCL) method is proposed for MRSI classification. Joint intramodal contrastive learning (IMCL) and CMCL are used to better mine multimodal feature representations during pretraining, and the IMCL and CMCL objectives are jointly optimized, whereby it encourages the learned representation to be semantically consistent within and between modalities simultaneously. Moreover, a simple but effective hybrid cross-modal fusion module (HCFM) is designed in the fine-tuning stage, which could better compactly integrate complementary information across these modalities for more accurate classification. Extensive experiments are taken on four benchmark datasets (i.e., Houston 2013, Augsburg, Germany; Trento, Italy; and Berlin, Germany), and the results show that the proposed method outperforms state-of-the-art methods.**

***Index Terms*— Cross-modal contrastive learning (CMCL), multimodal remote sensing image (MRSI) classification, self-supervised.**

I. Introduction

THE classification of remote sensing (RS) images plays an increasingly crucial role in geological exploration [1], environmental monitoring [2], urban planning [3], [4], etc. RS image classification is one of the critical techniques in the field of RS image processing. Recently, owing to advanced RS techniques and sensor techniques, different ground properties of the same geographical location of the Earth can be observed through multiple RS data, such as hyperspectral images (HSI) containing rich spectral reflectance information [5], [6], light detection and ranging (LiDAR) data providing ground elevation information [7], [8], and synthetic aperture radar (SAR) data, with an advantage of all-time and all-weather that provides the structure information of the ground surface [9]. However, it is difficult to exploit the potential of diverse RS data when using single RS data and inevitably encounters the bottleneck of RS image classification performance [10]. Compared with the classification of single-modal images, multimodal RS image (MRSI) provides more comprehensive information to represent ground properties by taking full advantage of complementary strengths, such as the HSI classification defect that HSI data make it challenging to distinguish ground objects with similar spectral responses is remedied by providing elevation information through LiDAR data [11], [12] (e.g., concrete roof and road). Moreover, the effect of inclement weather on HSI data and the serious speckle noise on SAR data is reduced by the fusion of SAR and HSI data [13].

Therefore, the hot issue in MRSI classification is how to better explore and integrate heterogeneous features. Traditional multimodal fusion algorithms have been applied to MRSI classification [14], [15], [16], [17], such as Pedergnana et al. [14] extracted fusion features to perform MRSI classification by calculating extended attribute profiles (EAPs) on HSI and LiDAR data. Liao et al. [15] proposed a generalized graph-based fusion strategy to fuse spectral information of HSI data and morphological profiles (MPs) of HSI and LiDAR data. However, the limited ability to represent multimodal heterogeneous features of traditional models is based on manual features. With the rapid development of deep learning (DL), convolutional neural networks (CNN) as a typical DL model, abundant MRSI classification methods based on CNN have been proposed. Such as multiscale interactive fusion network (MIFNet) [18], Co-CNN [19] that integrates HSI and LiDAR data in a cascade way, CCR-Net [20] that develops a novel cross-channel reconstruction (CCR) mechanism, deep hierarchical vision transformer (DHViT) fusion model [21], and joint CNN and morphological feature learning [22].

The existing mainstream methods can be roughly divided into two categories according to the integration strategy, i.e., feature-level and decision-level fusions. Feature-level fusion aims to integrate the extracted features of different modalities. Dong et al. [23] designed CNN-based multibranch feature extractors with self- and cross-guided attention to extract spectral, spatial, and elevation features for HSI and LiDAR data, then the three types of feature vectors were concatenated for classification. The main idea of decision-level fusion is to vote on multiple classifiers to obtain the best classification

Manuscript received 8 March 2023; revised 8 June 2023 and 3 July 2023; accepted 13 July 2023. Date of publication 19 July 2023; date of current version 7 August 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62171357 and 62276205; in part by the Science and Technology Innovation Team in Shaanxi Province of China under Grant 2020TD-017; in part by the Foundation of Key Laboratory of Aerospace Science and Industry Group of China Aerospace Science and Industry Corporation Limited (CASIC), China; in part by the Foundation of Intelligent Decision and Cognitive Innovation Center of State Administration of Science, Technology and Industry for National Defense, China; and in part by the Key Project of Hubei Provincial Natural Science Foundation under Grant 2020CFA001, China. *(Corresponding author: Shuyuan Yang.)*

The authors are with the School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: zxfeng@xidian.edu.cn; songll@stu.xidian.edu.cn; syyang@xidian.edu.cn; xinyu_zhang@stu.xidian.edu.cn; lchjiao@mail.xidian.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3296703

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:24:13 UTC from IEEE Xplore. Restrictions apply.

result. Zhang et al. [24] proposed an ensemble multiple kernel active learning framework based on the maximum disagreement query strategy for HSI and LiDAR data, and the final result was obtained by implementing decision-level fusion. In [19], a coupled CNN-based feature extractor was proposed, then the HSI and LiDAR data feature vectors were integrated by decision-level and feature-level fusions.

Despite the excellent performance gain in MRSI classification for these approaches, the following two issues remain.

1) *A Large Number of Labeled Instances for Comprehensive Feature Learning:* Considering the high cost and scarcity of labeled RS data, many DL-based MRSI classification methods driven by massive data fail to achieve the expected performance under limited instances. With the success of contrastive learning in computer vision (CV) [25], [26], [27], self-supervised contrastive learning has been shown to be effective in addressing this issue. Although recent work has investigated contrastive learning for RS image classification [28], [29], [30], [31], the existing contrastive learning-based methods encourage representation learning with invariance to a set of random augmentations for single-modal RS data. Therefore, these approaches do not consider the mining heterogeneous feature representation of MRSI, which can provide more comprehensive information to depict ground properties and improve classification performance.
2) *The Limitations of Existing Fusion Strategies:* Effective fusion of heterogeneous features is crucial to MRSI classification. The existing fusion strategies based on concatenation and convolution operations focus on the fusion of heterogeneous features in local regions [19], [20], [22], and have limitations in dealing with the global dependencies of MRSI. And the other methods based on a fusion of fully connected layers inevitably have an oversized number of parameters [11], [32], [33]. In addition, the direct stacking of higher-level features may be influenced by the Hughes phenomenon. These factors restrict the performance of the MRSI classification. Cross-attention based on the self-attention mechanism has been demonstrated to capture global dependencies of MRSI and adaptively integrate heterogeneous features through contextual awareness [18], [21]. Current methods typically concentrate on either local or global feature fusion, with limited research on how to effectively combine both. Therefore, effectively fusing different modalities remains a challenge for MRSI classification.

In our work, taking inspiration from the success of contrastive learning, [25], [28], [29], [30], we propose an effective self-supervised cross-modal contrastive learning (CMCL) method for MRSI classification. First, 3-D patches centered on target pixels are sampled from the RS image through a sliding window. Then the intramodal patches are sent to the respective contrastive learning framework to learn the intramodal feature representation. Moreover, a novel CMCL framework is presented to learn discriminative and complementary features, which could capture the correspondence between MRSI and yield better representational capabilities. More importantly, an elegant integration of cross-attention and convolution is achieved called hybrid cross-modal fusion module (HCFM) in the fine-tuning stage, in which local aware fusion module (LAFM) and global aware fusion module (GAFM) are included, which further integrates heterogeneous features in a more compact and efficient way, resulting in powerful fusion feature representation for classification. The main contributions are summarized as follows.

1) A novel self-supervised CMCL framework is proposed for MRSI classification with limited labeled instances. To the best of our knowledge, it is the first work on MRSI classification.
2) The joint intramodal contrastive learning (IMCL) and CMCL to relate the compositional patterns in MRSI. The correspondence between multimodal data is further captured while learning the intramodal feature representation.
3) A simple but effective HCFM is introduced, which can simultaneously fuse local and global features to better compactly integrate cross-modal complementary information and capture global dependencies of heterogeneous features, and better MRSI classification performance is achieved.

The rest of this article is organized as follows. Related work regarding contrastive learning and MRSI classification is introduced in Section II. Section III gives the details of the proposed method. Extensive comparative experiments and ablation studies are conducted in Section IV, and the conclusion is presented in Section V.

II. Related Work

A. Self-Supervised Contrastive Learning on RS Image Classification

Self-supervised learning learns data representations from unlabeled datasets, which has succeeded in CV tasks [25], [26], [27]. As a popular self-supervised learning strategy, contrastive learning aims to construct potential space so that similar samples are close together and dissimilar ones are far away. Numerous outstanding contrastive learning algorithms, such as simple framework for contrastive learning of visual representations (SimCLR) [25], bootstrap your own latent (BYOL) [26], and momentum contrast (MoCo) [27], have been presented in the field of CV.

Several works based on contrastive learning RS image classification have been conducted. Liu et al. [28] constructed two views of HSI by principal component analysis (PCA) of different spectral bands, then maximized the consistency loss between different enhanced views of the same data using a deep residual network. Zhu et al. [29] proposed a self-supervised multiscale pixel-wise contrastive learning model and a lightweight feature extraction network was devised and inserted into the contrastive learning framework. Xu et al. [34] first utilized transformations to obtain two feature views of the same HSI data, then a spectral–spatial feature extraction network was proposed to project the two views into the embedding space, and finally a contrastive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:24:13 UTC from IEEE Xplore. Restrictions apply.

loss function is applied to align the two projected features. Contrary to the existing RS image classification works based on single-modal contrastive learning, we jointly optimize IMCL and CMCL objectives, which could learn not only the intramodal feature representation but also capture multimodal correspondences, yielding a better classification performance under limited samples.

B. MRSI Classification

It is difficult to exploit the potential of multimodal heterogeneous features using single-modal RS data, such as HSI, LiDAR, and SAR data, and inevitably encounters the bottleneck of RS image classification performance. Combined multimodal data can provide more diverse information to depict the ground properties and has shown significant advantages and potential in RS image classification. Some works have been explored for collaborative classification for MRSI.

1) *Multimodal Fusion Classification of HSI and LiDAR Data*: HSI containing abundant spectral and spatial information and LiDAR data recording elevation information using complementary strengths for synergistic classification is a meaningful paradigm. Initial lines of work mainly utilized MPs and EAPs [14], [15], [35], [36]. The recent lines of work have applied CNN to multimodal fusion classification of HSI and LiDAR data, obtaining classification performance that is superior to single-modal data. Chen et al. [32] proposed a feature fusion network based on CNN, and multimodal heterogeneous features were fused by a deep neural network (DNN) for classification. Wu et al. [20] proposed a CNN framework with a CCR module called CCR-Net, and CCR-Net learns more compact fused representations of heterogeneous features through cross-modal reconstruction strategies. Self-attention mechanism-based transformer can handle long-range dependencies of input sequence [37], [38]. Several approaches have been explored to integrate multimodal features based on a novel transformer framework. Xue et al. [21] designed a novel multibranch DHViT framework for the joint classification of HSI and LiDAR data, moreover, the heterogeneous features were fused via cross-attention mechanism. Ding et al. [39] proposed a global–local transformer network for the joint classification of HSI and LiDAR data, and multiscale feature fusion and probabilistic decision fusion were also applied to further improve the classification performance.
2) *Multimodal Fusion Classification of HSI and SAR Data*: SAR data benefits from the all-time and all-weather advantage, reducing the impact of bad weather on HSI. Therefore, the joint classification of HSI and SAR data is also a meaningful paradigm. Recent works have explored DL for multimodal fusion classification of heterogeneous features from HSI and SAR data. Li et al. [40] proposed an asymmetric feature fusion network (AsyFFNet) for HSI and SAR image classification, the weight-share residual blocks are designed to extract features, and the feature calibration module is devised to enhance the discrimination capability. Wang et al. [18] extracted HSI and SAR data features using a two-branch structure based on multiscale information extraction, then a global dependence fusion module was designed to integrate two types of features.

In contrast to the existing methods, our proposed method yields a better representation capability by jointing intra and cross-modal self-supervised contrastive learning, and HCFM can better compactly integrate cross-modal information and capture global dependencies of multimodal heterogeneous features. Moreover, the proposed method is a novel generalized MRSI classification framework that focuses on learning RS feature representation without being limited to fixed modalities.

III. Proposed Method

In this section, we will introduce the details of the proposed self-supervised CMCL framework, including IMCL and CMCL, in the pretraining stage, and HCFM in the fine-tuning stage. Finally, the overall objective is presented. The overview of the proposed method is shown in Fig. 1.

A. Intramodal Contrastive Learning

Inspired by the success of contrastive learning [25], [26], [27], we formulate the IMCL to learn the feature representation within modalities. Given a MRSI dataset $D = \{(H_i, M_i)\}_{i=1}^{|D|}$ with $H_i \in R^{S\times S\times H}$ and $M_i \in R^{S\times S\times W}$. H_i and M_i are patches centered on the target pixel sampled from the same region of the MRSI by means of the sliding window, and the sliding window size is $S \times S$.

First, given an input modal H_i, we construct its augmented versions H_i^{aug1} and H_i^{aug2} by performing feature transformations T, and we also construct augmented versions M_i^{aug1} and M_i^{aug2} for another modal M_i. T represents random transformations such as rotation and flip. Second, the feature extractor f_{θ_h} and $f_{\theta m}$ maps H_i^{aug1} and H_i^{aug2}, M_i^{aug1} and M_i^{aug2} to the feature embedding space, respectively. Then H_i^{aug1} and H_i^{aug2} are projected to a new embedding space by using the projection head g_{φ_h}, and similarly maps M_i^{aug1} and M_i^{aug2} to a new embedding space using projection head g_{φ_m}. The network configuration of the feature extractor and projection head is shown in Table I. The process of feature mapping can be formulated as

$$z_i^{\text{aug1}} = g_{\varphi_h}\left(f_{\theta_h}\left(H_i^{\text{aug1}}\right)\right) \tag{1}$$

$$z_i^{\text{aug2}} = g_{\varphi_h}\left(f_{\theta_h}\left(H_i^{\text{aug2}}\right)\right) \tag{2}$$

$$h_i^{\text{aug1}} = g_{\varphi_m}\left(f_{\theta_m}\left(M_i^{\text{aug1}}\right)\right) \tag{3}$$

$$h_i^{\text{aug2}} = g_{\varphi_m}\left(f_{\theta_m}\left(M_i^{\text{aug2}}\right)\right) \tag{4}$$

where z_i^{aug1} and z_i^{aug2} denote the final projection vectors of H_i^{aug1} and H_i^{aug2}, respectively, and similarly h_i^{aug1} and h_i^{aug2} represent the final projection vectors of M_i^{aug1} and M_i^{aug2}, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:24:13 UTC from IEEE Xplore. Restrictions apply.

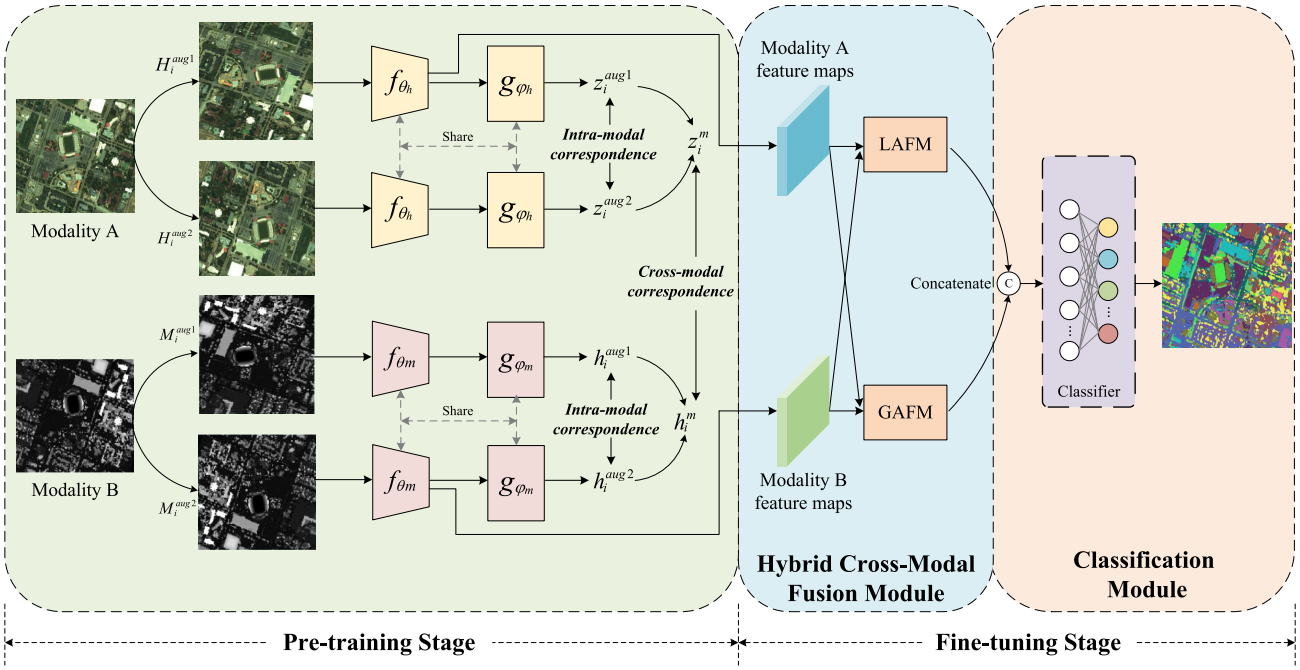


Fig. 1. Overall architecture of the proposed method. The pretraining phase consists of two branches, namely the intramodal correspondence of the two modalities by introducing contrastive loss. Moreover, cross-modal correspondence is also established. In the fine-tuning phase, feature extractors for both modalities as the backbone of the classification task, and heterogeneous features are further fused by HCFM and then feeds them into the classification module.

TABLE I
NETWORK CONFIGURATION OF THE FEATURE EXTRACTOR AND PROJECTION HEAD FOR TWO MODALITIES

Modality	Feature Extractor			Projection Head	
	f_{θ_h}			g_{φ_h}	
	Block 1	Block 2	Block 3	Block 4	Block 5
Modality A	3×3 Conv	3×3 Conv	3×3 Conv	FC	FC
	BN	BN	BN	-	-
	ReLU	ReLU	ReLU	-	-
	f_{θ_m}			g_{φ_m}	
	Block 1	Block 2	Block 3	Block 4	Block 5
Modality B	3×3 Conv	3×3 Conv	3×3 Conv	FC	FC
	BN	BN	BN	-	-
	ReLU	ReLU	ReLU	-	-

For the positive pair of examples z_i^{aug1} and z_i^{aug2} of modality A, the contrastive loss function can be formulated as follows:

$$l_h(i, \text{aug1}, \text{aug2}) = -\log \frac{\exp\left(s\left(z_i^{\text{aug1}}, z_i^{\text{aug2}}\right)/\tau\right)}{\sum_{k=1}^{2N} \Gamma_{[k\neq i]} \exp\left(s\left(z_i^{\text{aug1}}, z_k\right)/\tau\right)} \tag{5}$$

where $\Gamma_{[k\neq i]} \in \{0, 1\}$ denotes the indicator function which equals to 1 if $k \neq i$, $s(\cdot)$ denotes the cosine similarity function, and N is the mini-batch size. τ denotes a temperature co-efficient, the default is set to 0.1. The modality A contrastive learning loss function for a mini-batch size can be defined as follows:

$$\mathcal{L}_{\text{imcl}}^{h} = \frac{1}{2N} \sum_{i=1}^{\text{N}} \left[l_h(i, \text{aug1}, \text{aug2}) + l_h(i, \text{aug2}, \text{aug1})\right] \tag{6}$$

similarly, for the positive pair of examples h_i^{aug1} and h_i^{aug2} of modality B, the contrastive loss function can be defined as follows:

$$l_m(i, \text{aug1}, \text{aug2}) = -\log \frac{\exp\left(s\left(h_i^{\text{aug1}}, h_i^{\text{aug2}}\right)/\tau\right)}{\sum_{k=1}^{2N} \Gamma_{[k\neq i]} \exp\left(s\left(h_i^{\text{aug1}}, h_k\right)/\tau\right)}. \tag{7}$$

The modality B contrastive learning loss function for a mini-batch size can be formulated as follows:

$$\mathcal{L}_{\text{imcl}}^{m} = \frac{1}{2N} \sum_{i=1}^{N} \left[l_m(i, \text{aug1}, \text{aug2}) + l_m(i, \text{aug2}, \text{aug1})\right]. \tag{8}$$

B. Cross-Modal Contrastive Learning

In addition to the IMCL, we introduce a novel CMCL objective to learn discriminative features, which capture the correspondence between MRSI and yield better representation capability.

First, the mean of z_i^{aug1} and z_i^{aug2} is calculated as z_i^m, and the mean h_i^m of h_i^{aug1} and h_i^{aug2}. Then the loss function for the positive pair of examples z_i^m and h_i^m as follows:

$$l_c\left(i, z_i^m, h_i^m\right) = -\log \frac{\exp\left(s\left(z_i^m, h_i^m\right)/\tau\right)}{\sum_{\substack{k=1 \\ k\neq i}}^{N} \exp\left(s\left(z_i^m, z_k^m\right)/\tau\right) + \sum_{k=1}^{N} \exp\left(s\left(z_i^m, h_k^m\right)/\tau\right)} \tag{9}$$

where $s(\cdot)$, N, and τ have the same parameters as in (7). The CMCL loss function for a mini-batch size can be formulated as follows:

$$\mathcal{L}_{\text{cmcl}} = \frac{1}{2N} \sum_{i=1}^{N} \left[l_c\left(i, z^m, h^m\right) + l_c\left(i, h^m, z^m\right)\right]. \tag{10}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:24:13 UTC from IEEE Xplore. Restrictions apply.

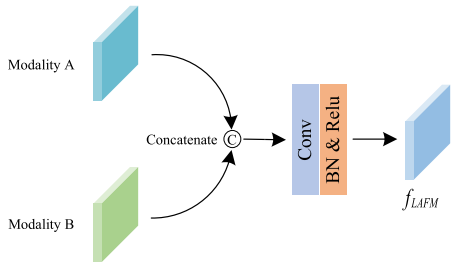


Fig. 2. Details of the LAFM.

The CMCL maximizes the similarity of z_i^m and h_i^m, this enforces the model to learn from harder positive and negative RS image samples, thus yield better representation capability.

C. Hybrid Cross-Modal Fusion Module

After the pretraining stage of IMCL and CMCL, the feature extractor yields better representation capability. To further fuse MRSI information, we propose a novel HCFM in the fine-tuning stage, which includes LAFM and GAFM, in order to integrate cross-modal information and construct global dependencies of MRSI features more compactly, respectively. Once the IMCL and CMCL are completed, the backbone network f_{θ_h} and f_{θ_m} can be an effective feature extractor. Therefore, we concatenate HCFM behind the backbone network to further complete the fusion and classification.

Given the outputs f_a and f_b of backbone network f_{θ_h} and f_{θ_m}, respectively. There has been complementary information in the same place across different channels of f_a and f_b. Therefore, a convolution layer with 1×1 kernels is applied as an LAFM to integrate cross-modal information. Fig. 2 illustrates the details of LAFM, which is given by

$$f_{\text{LAFM}} = \sigma\left(w\left[f_a, f_b\right] + b\right) \tag{11}$$

where w and b are the learnable weights and bias of 1×1 filter, and σ is the activation function ReLU. $[f_a, f_b]$ denotes the concatenating of f_a and f_b in the channel dimension.

In contrast to LAFM, GAFM is achieved by an interactive cross-modal attention mechanism to mine the multimodal global dependencies. Fig. 3 illustrates the details of GAFM. First, the attention maps of $f^a_{\text{attention}}$ and $f^b_{\text{attention}}$ are generated from f_a and f_b, respectively. It can be formulated as follows:

$$f^a_{\text{attention}} = \text{softmax}\left(\left(f_{\text{conv}}(f_a)\right)^{\text{T}} f_{\text{conv}}(f_a)\right) \tag{12}$$

$$f^b_{\text{attention}} = \text{softmax}\left(\left(f_{\text{conv}}(f_b)\right)^{\text{T}} f_{\text{conv}}(f_b)\right) \tag{13}$$

where f_{conv} denotes convolution layer with 1×1 kernels, f_a and f_b refers to the same parameters as (11). Second, cross attention between MRSI is established, it is given by

$$f^a_{\text{GAFM}} = \left(f_{\text{conv}}(f_b) \otimes f^a_{\text{attention}}\right) \oplus f_a \tag{14}$$

$$f^b_{\text{GAFM}} = \left(f_{\text{conv}}(f_a) \otimes f^b_{\text{attention}}\right) \oplus f_b. \tag{15}$$

Complementary guidance relations between modalities A and B are captured by f^a_{GAFM} and f^b_{GAFM}, respectively. Moreover, the feature maps f^a_{GAFM} and f^b_{GAFM} with cross-attention enhancement are concatenated to obtain a more powerful representation of fused features, it is given by

$$f_{\text{GAFM}} = \left[f^a_{\text{GAFM}}, f^b_{\text{GAFM}}\right] \tag{16}$$

where $[,]$ refers to the same meaning as (11).

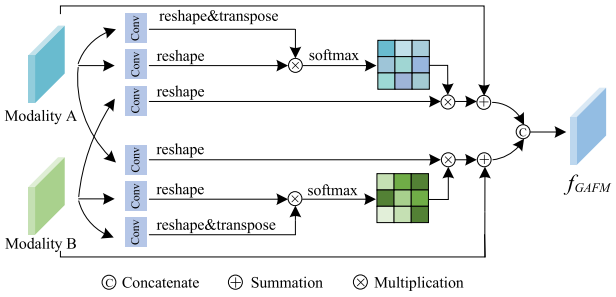


Fig. 3. Details of the GAFM.

D. Classification Module

In the fine-tuning stage, the outputs of LAFM and GAFM are concatenated together to obtain a more powerful representation of the complementary fusion features, it can be formulated as follows:

$$f_{\text{output}} = \left[f_{\text{LAFM}}, f_{\text{GAFM}}\right] \tag{17}$$

where $[,]$ refers to the same meaning as (11). Then f_{output} is fed to the classification module consisting of fully connected layers, the classification loss can be computed by

$$\mathcal{L}_{\text{sup}} = -\frac{1}{N_{\text{sup}}} \sum_{i=1}^{N_{\text{sup}}} y_i \log y_i' \tag{18}$$

where N_{sup} denotes the training samples with labels in each mini-batch during the fine-tuning phase. y_i represents the ground truth of i th training sample and y_i' is predictive label.

E. Overall Objective

Finally, the total loss function in the self-supervised pretraining stage can be given by

$$\mathcal{L}_{\text{self}} = \mathcal{L}^h_{\text{imcl}} + \mathcal{L}^m_{\text{imcl}} + \mathcal{L}_{\text{cmcl}} \tag{19}$$

where $\mathcal{L}^h_{\text{imcl}}$ and $\mathcal{L}^m_{\text{imcl}}$ are IMCL loss function, and $\mathcal{L}_{\text{cmcl}}$ is CMCL loss function. The classification loss function in the fine-tuning stage is $\mathcal{L}_{\text{sup}}$.

IV. Experimental Results and Analysis

A. Datasets Description

1) *HSI-LiDAR Houston 2013 Dataset:* The dataset was obtained by the ITRES CASI-1500 sensor, which covered the Houston University campus and its adjacent area. The HSI has 144 spectral bands with wavelengths ranging from 0.38 to 1.05 μm. Both the spatial size of HSI and LiDAR data is 349×1905 with a spatial resolution of 2.5 m. The dataset has a total of 15 land cover classes.
2) *HSI-SAR Augsburg Dataset:* The dataset was collected in Augsburg, Germany, and the HSI and dual-polarimetric SAR (PolSAR) data were captured by the HySpex sensor and the Sentinel-1 sensor, respectively. Both the spatial size is 332×485 with a spatial resolution of 30 m. The HSI contains 180 spectral bands in the wavelength range

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:24:13 UTC from IEEE Xplore. Restrictions apply.

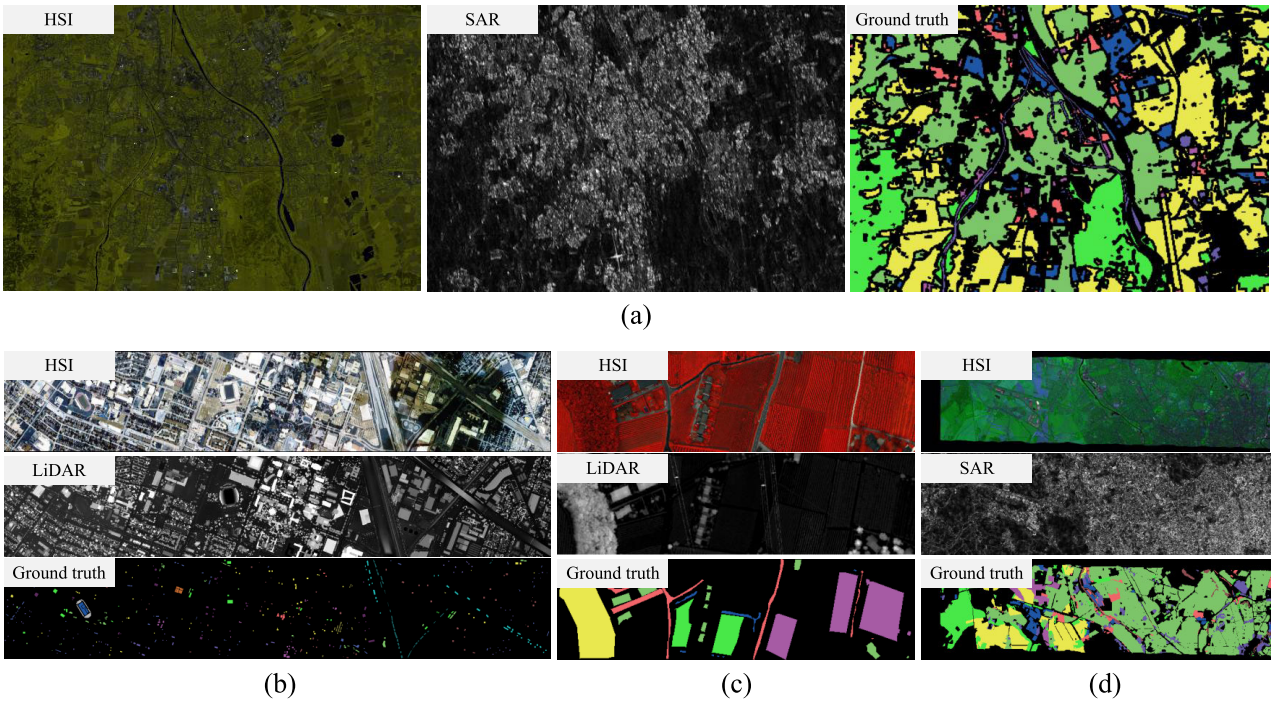


Fig. 4. MRSI datasets. (a) Augsburg dataset. (b) Houston 2013 dataset. (c) Trento dataset. (d) Berlin dataset.

of 0.4–2.5 μm, and the SAR contains four features. Seven land cover classes are contained in this dataset.

3) *HSI-LiDAR Trento Dataset:* The dataset was acquired by the AISA Eagle sensor and Optech ALTM 3100EA sensor covering a rural area in the south of Trento, Italy. The HSI contains 63 spectral bands ranging from 0.42 to 0.99 μm. Both the spatial size is 166 $\times$ 600 with a spatial resolution of 1 m, and there have been six land cover classes.
4) *HSI-SAR Berlin Dataset:* The dataset covered the geomorphic makeup of both urban and rural areas surrounding Berlin. The HSI is sourced from EnMAP data synthesized from the HyMap HSI data, while SAR data is obtained from the European Space Agency's (ESA) Sentinel-1 dual-Pol (VV–VH) single-look complex (SLC) product. The size of HSI is 797 $\times$ 220 pixels, with 244 spectral bands covering wavelengths from 0.4 to 2.5 μm and a spatial resolution of 30 m. The SAR data comprises an image of 1723 $\times$ 476 pixels and a spatial resolution of 13.89 m, and there have been eight land cover classes. The HSI was processed using the nearest neighbor interpolation algorithm to align its size with that of the SAR image.

The visualized images of the four datasets are shown in Fig. 4, including the false color image of HSI, the gray-scale image of LiDAR, and the SAR image.

B. Experimental Setting

1) Evaluation Metrics: In the experiments, four quantitative evaluation indices [i.e., accuracy for each category, overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ)] are applied to evaluate the MRSI classification performance.

2) Implementation Details: All experiments are implemented on the computer using the PyTorch framework with NVIDIA TITAN V GPU with 12-GB memory. Adam optimizer [47] is applied to optimize the proposed model. For our proposed model, in the self-supervised contrastive learning pretraining stage, the batch size is set to 64, the learning rate is specified as 0.001, and 50 epochs are executed. For four datasets, 50% of the unlabeled samples are selected as training samples. In the fine-tuning procedure, the batch size is set as 16, and the learning rate is set to 0.0005, and also 50 epochs are executed. For four datasets, ten labeled samples for each class are randomly selected as training samples and the rest as testing samples in the experiments. The details of the training samples and testing samples are listed in Table II. Moreover, PCA is first applied to process the original HSI data of these four datasets to reduce the amount of computation, and the empirical setting of spectral dimensions after dimension reduction by PCA is 30 [48], and the patch size of MRSI is empirically set as 11 [31].

C. Comparison to State-of-the-Art Networks

To assess the effectiveness of the proposed model, several representative state-of-the-art methods are selected for the comparison, including four single-modal classification methods-based HSI, i.e., the conventional methods (i.e., support vector machines (SVM) [41] with radial basis function (RBF) kernel) and the DL-based methods (i.e., contextual deep CNN (CDCNN) [42], spectral–spatial residual network (SSRN) [43], and double-branch multiattention mechanism network (DBMA) [44]), and a method in the CV field (i.e., SimCLR [25]), and also including five MRSI classification methods (i.e., Co-CNN [19], multimodal deep learning meets remote sensing (MDL-RS) [45], CCR-Net [20], MIFNet [18], and self-supervised feature representation (SSFR) [46]). Note that these methods use the original parameters from the article, and as in the fine-tuning stage of our proposed model,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:24:13 UTC from IEEE Xplore. Restrictions apply.

TABLE II

LAND COVER CLASSES, AND THE NUMBER OF LABELED TRAINING SAMPLES AND TESTING SAMPLES FOR FOUR DATASETS, INCLUDING HOUSTON 2013, AUGSBURG, TRENTO, AND BERLIN DATASETS

Class No.	Houston 2013			Augsburg			Trento			Berlin		
	Class Name	Training	Testing	Class Name	Training	Testing	Class Name	Training	Testing	Class Name	Training	Testing
1	Healthy Grass	10	1241	Forest	10	13497	Apple	10	4024	Forest	10	54944
2	Stressed Grass	10	1244	Residential Area	10	30319	Buildings	10	2893	Residential Area	10	268632
3	Synthetic Grass	10	687	Industrial Area	10	3841	Ground	10	469	Industrial Area	10	19556
4	Tree	10	1234	Low Plants	10	26847	Woods	10	9113	Low Plants	10	59272
5	Soil	10	1232	Allotment	10	565	Vineyard	10	10491	Soil	10	17416
6	Water	10	315	Commercial Area	10	1635	Roads	10	3164	Allotment	10	13295
7	Residential	10	1258	Water	10	1520				Commercial Area	10	24814
8	Commercial	10	1234							Water	10	6662
9	Road	10	1242									
10	Highway	10	1217									
11	Railway	10	1225									
12	Parking Lot1	10	1223									
12	Parking Lot1	10	1223									
13	Parking Lot2	10	459									
14	Tennis Court	10	418									
15	Running Track	10	650									
	Total	150	14879	Total	70	78224	Total	60	30154	Total	80	464591

TABLE III

OA, AA, κ, AND THE ACCURACY OF PER CLASS FOR DIFFERENT METHODS ON HOUSTON 2013 DATASET IN WHICH THE BEST RESULTS ARE DENOTED IN BOLD

Class	SVM [41]	CDCNN [42]	SSRN [43]	DBMA [44]	Co-CNN [19]	MDL-RS [45]	CCR-Net [20]	MIFNet [18]	SimCLR [25]	SSFR [46]	Proposed
1	88.65±4.32	81.77±9.50	83.84±5.78	88.33±4.24	86.51±2.81	90.62±5.56	91.34±3.21	90.07±5.59	94.01±3.01	**95.65**±3.49	93.01±1.81
2	91.38±6.54	89.77±10.7	**94.26**±5.24	92.36±4.33	92.41±4.31	88.93±7.79	89.08±5.98	79.61±8.76	74.78±3.88	86.96±4.61	93.38±1.24
3	87.88±11.2	88.05±15.7	99.64±0.43	99.94±0.11	97.87±1.56	97.86±1.37	96.50±1.99	92.64±5.21	99.18±0.95	99.10±1.54	**99.98**±0.04
4	96.09±3.53	94.96±5.69	97.07±2.25	94.66±8.99	89.54±7.52	**98.81**±0.84	87.36±11.5	85.55±6.35	95.12±1.43	88.50±5.65	87.01±1.87
5	90.91±2.62	95.36±2.75	93.15±4.83	93.91±3.11	**99.54**±0.46	98.67±1.59	99.09±0.55	98.56±1.59	91.31±3.96	94.37±3.86	95.25±1.33
6	93.99±5.88	85.34±6.97	94.97±8.88	95.22±3.26	91.14±7.51	85.29±7.30	90.79±5.04	94.41±5.43	**96.30**±4.41	95.57±3.22	89.07±3.13
7	67.72±9.40	74.40±4.42	75.90±10.1	78.17±12.6	64.40±6.54	75.43±1.18	68.55±10.1	70.91±11.3	**88.21**±4.03	83.12±5.96	86.84±2.50
8	66.75±10.9	81.29±9.67	80.49±21.4	**94.07**±6.12	73.39±5.59	75.93±6.07	73.58±13.5	68.95±11.8	83.36±4.17	78.37±6.49	75.72±3.21
9	62.88±9.74	70.61±9.83	66.63±7.93	75.42±8.31	61.79±10.8	70.55±7.16	60.99±12.1	69.20±9.33	72.69±5.04	78.63±5.82	**80.50**±3.61
10	59.57±7.49	61.68±9.19	69.39±12.3	70.02±10.7	83.54±8.28	74.11±3.85	84.82±8.56	73.33±15.8	62.76±4.90	66.97±4.76	**89.74**±1.58
11	58.80±6.83	65.33±9.44	77.09±6.22	61.67±11.1	**88.71**±4.97	78.44±13.8	83.55±8.86	75.13±11.3	77.67±3.95	88.57±6.03	84.34±4.03
12	59.63±4.29	74.23±5.62	73.57±9.88	75.75±8.49	**86.31**±4.70	74.08±13.5	74.75±16.9	78.54±9.88	63.02±6.18	78.74±5.04	72.04±4.28
13	31.88±10.3	80.28±8.39	93.11±2.99	77.75±13.8	89.84±1.96	**94.84**±0.37	88.21±5.25	87.69±6.89	60.26±4.84	62.32±14.1	81.35±7.85
14	79.28±8.88	73.72±11.9	85.64±16.7	98.46±3.06	**99.88**±0.22	96.57±4.68	98.51±1.26	98.01±3.54	96.55±2.87	91.99±5.16	95.57±2.67
15	99.26±0.53	88.54±6.66	95.90±1.96	92.70±4.77	99.23±0.85	99.12±0.59	89.69±29.8	99.07±1.57	98.76±1.26	98.46±1.28	**99.50**±0.64
OA	74.36±0.02	78.32±1.74	81.79±4.77	82.16±3.27	84.89±1.21	84.82±0.94	83.25±2.14	81.62±3.09	80.80±1.17	84.15±1.03	**86.90**±0.34
AA	75.64±0.01	80.36±1.63	85.38±3.84	85.90±2.11	86.94±1.14	86.62±0.99	85.12±2.36	84.11±2.63	83.60±0.61	85.82±0.69	**88.22**±0.54
$\kappa\times100$	72.29±0.02	76.58±1.87	80.31±5.15	80.72±3.53	83.67±1.31	83.60±1.01	81.89±2.32	80.14±3.32	79.26±1.26	82.88±1.11	**85.83**±0.37

TABLE IV

OA, AA, κ, AND THE ACCURACY OF PER CLASS FOR DIFFERENT METHODS ON AUGSBURG DATASET IN WHICH THE BEST RESULTS ARE DENOTED IN BOLD

Class	SVM [41]	CDCNN [42]	SSRN [43]	DBMA [44]	Co-CNN [19]	MDL-RS [45]	CCR-Net [20]	MIFNet [18]	SimCLR [25]	SSFR [46]	Proposed
1	69.54±9.00	69.40±8.05	87.12±6.04	86.33±7.01	**96.58**±2.06	93.96±4.52	95.44±1.77	87.09±13.58	65.08±6.37	87.46±2.62	93.63±1.32
2	70.70±6.47	73.05±3.51	91.39±4.36	91.24±5.27	69.57±11.9	59.84±17.2	55.13±13.6	52.82±11.06	86.86±3.93	94.90±1.20	**95.42**±0.53
3	23.76±7.96	24.96±7.87	40.69±11.0	46.63±17.4	45.54±12.8	42.73±21.6	37.51±15.6	56.32±11.68	28.12±5.99	31.33±11.6	**60.49**±1.40
4	77.43±4.23	73.21±5.73	95.27±4.81	97.10±2.05	75.86±11.2	82.79±5.79	70.11±14.9	73.21±13.3	85.85±4.12	**99.07**±0.39	97.70±0.40
5	4.13±0.73	4.80±1.26	11.67±5.24	11.75±4.75	81.45±14.1	**81.94**±9.72	78.67±13.5	72.90±8.98	7.01±1.66	9.72±4.01	5.83±1.09
6	10.08±2.79	12.68±4.00	16.90±2.80	19.42±4.13	35.38±14.2	**43.62**±20.7	33.81±12.5	35.54±10.11	9.49±1.70	14.35±2.58	28.23±1.37
7	19.51±9.12	16.55±4.23	35.90±17.8	24.95±14.5	51.92±6.36	56.84±11.1	59.08±8.12	**61.32**±1.90	13.31±4.77	37.49±11.9	33.95±5.21
OA	49.53±3.91	49.59±4.24	74.45±5.36	73.09±4.91	74.24±7.03	72.53±6.73	66.16±6.91	65.85±6.45	55.89±2.99	74.36±4.01	**79.22**±2.11
AA	39.31±2.45	39.24±2.04	54.14±3.31	53.92±2.71	65.19±3.50	**65.96**±3.40	61.39±3.00	62.75±1.88	42.24±1.52	53.48±2.37	59.32±0.60
$\kappa\times100$	37.09±4.35	36.97±4.20	66.19±6.38	64.69±5.60	65.91±8.18	64.10±7.46	56.35±6.85	56.61±6.87	44.66±3.02	66.36±4.52	**72.09**±2.60

ten labeled samples for each class are randomly selected as training samples for these methods.

1) Quantitative Evaluation: The OA, AA, κ, and the accuracy per class for different methods are reported in Tables III–VI. Each experiment is executed five times, where the mean the standard deviation of each index are reported in the experiments.

According to the results, first, due to the excellent feature extraction capability of DL, all DL-based methods achieve better classification performance compared to the traditional method on four datasets. Moreover, deeper models (i.e., SSRN and DBMA) can yield higher classification accuracies than simple models (i.e., CDCNN), such as the OA of SSRN and DBMA over 25% higher than CDCNN on the Augsburg dataset, the OA of SSRN and DBMA is more than 8% higher than that of CDCNN on the Berlin dataset, and the reason is deeper models have the powerful ability in extracting stronger discriminative features.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:24:13 UTC from IEEE Xplore. Restrictions apply.

TABLE V

OA, AA, κ, AND THE ACCURACY OF PER CLASS FOR DIFFERENT METHODS ON TRENTO DATASET IN WHICH THE BEST RESULTS ARE DENOTED IN BOLD

Class	SVM [41]	CDCNN [42]	SSRN [43]	DBMA [44]	Co-CNN [19]	MDL-RS [45]	CCR-Net [20]	MIFNet [18]	SimCLR [25]	SSFR [46]	Proposed
1	49.32±4.87	71.60±8.36	90.46±6.19	91.84±3.14	91.40±6.61	96.61±1.91	**98.90**±0.83	84.77±25.3	67.47±6.55	87.52±1.09	98.29±1.87
2	68.37±6.99	75.78±2.52	76.53±4.54	78.59±3.61	88.62±5.74	92.02±5.92	96.34±2.72	92.45±4.84	75.78±4.61	**97.64**±0.60	96.42±1.95
3	73.67±4.10	68.02±13.4	93.50±6.16	88.57±6.21	95.18±2.85	**95.30**±2.21	89.57±7.33	91.40±11.5	64.58±10.2	81.52±6.50	89.18±2.67
4	94.06±3.30	93.41±5.27	99.60±0.15	99.63±0.08	99.62±0.43	99.93±0.11	99.34±0.92	97.26±7.89	82.56±4.37	**99.95**±0.05	99.94±0.04
5	88.00±4.94	95.76±1.55	98.41±0.77	97.74±1.23	**99.21**±1.00	92.37±6.49	96.92±3.77	92.21±4.69	88.37±4.37	99.41±0.13	98.95±1.69
6	69.90±8.33	81.10±2.17	83.65±8.37	82.78±6.73	**94.12**±1.89	91.41±4.54	89.79±2.84	85.26±11.7	76.70±4.35	91.66±0.74	89.33±4.10
OA	75.46±1.67	87.32±2.29	93.60±1.86	93.75±1.13	96.68±1.23	95.13±2.11	97.00±1.21	92.02±7.62	79.73±2.58	96.44±0.34	**97.62**±0.62
AA	73.89±1.42	80.95±3.27	90.36±2.72	89.86±1.81	94.69±1.71	94.61±1.38	95.14±1.20	90.56±8.95	75.91±2.57	92.95±1.21	**95.35**±0.87
$\kappa\times100$	68.26±2.02	83.13±3.03	91.50±2.46	91.68±1.49	95.56±1.67	93.56±2.74	96.01±1.58	89.49±9.91	73.21±3.32	95.27±0.45	**96.84**±0.82

TABLE VI

OA, AA, κ, AND THE ACCURACY OF PER CLASS FOR DIFFERENT METHODS ON BERLIN DATASET IN WHICH THE BEST RESULTS ARE DENOTED IN BOLD

Class	SVM [41]	CDCNN [42]	SSRN [43]	DBMA [44]	Co-CNN [19]	MDL-RS [45]	CCR-Net [20]	MIFNet [18]	SimCLR [25]	SSFR [46]	Proposed
1	61.57±8.81	69.66±6.65	72.42±9.27	79.46±5.81	**84.04**±3.36	79.51±2.80	78.00±1.68	80.22±3.40	70.62±10.3	73.10±6.70	73.26±11.0
2	90.25±1.72	88.88±3.27	90.37±2.56	**93.16**±0.76	46.73±13.5	52.38±16.0	44.78±6.71	56.01±9.07	86.62±2.46	90.41±1.72	86.46±3.93
3	24.03±6.76	34.43±6.56	32.03±10.0	29.65±6.32	43.82±8.93	43.50±11.6	37.06±13.0	35.91±5.75	26.82±5.21	31.92±7.83	**49.91**±9.89
4	66.29±6.13	75.26±3.35	81.24±3.02	83.29±3.23	78.95±2.56	66.13±11.0	70.21±6.07	78.02±4.96	85.85±4.83	81.45±5.61	**89.51**±3.07
5	57.73±7.65	53.18±12.3	67.06±9.32	65.48±8.82	81.52±13.9	**87.23**±3.99	86.84±5.42	85.92±6.64	65.06±9.25	61.52±7.88	85.67±11.1
6	9.60±1.43	12.12±2.38	11.41±2.06	16.42±3.86	83.45±3.21	67.22±18.8	73.34±4.84	**83.60**±4.61	13.34±2.71	13.35±2.89	15.28±4.37
7	19.04±9.25	14.67±3.68	22.60±5.27	21.91±4.21	32.85±5.60	38.82±17.8	**39.95**±7.99	36.85±9.08	20.87±2.19	27.26±9.18	25.12±6.55
8	40.78±21.7	19.59±8.45	43.12±17.6	22.53±2.69	**75.90**±7.95	65.95±9.98	70.36±8.28	71.51±10.8	27.92±7.07	39.98±19.6	31.95±17.1
OA	47.12±4.17	48.00±4.31	56.10±7.34	60.15±2.48	57.16±7.70	58.17±9.52	54.13±2.72	61.95±5.16	56.30±6.21	60.42±5.66	**67.35**±2.83
AA	46.16±2.95	45.97±1.78	52.53±5.74	51.49±0.98	65.91±2.51	62.59±3.53	62.57±1.46	**66.01**±1.48	49.64±1.13	52.37±4.70	57.15±2.72
$\kappa\times100$	35.74±2.74	36.61±3.07	44.16±6.85	48.28±2.11	45.56±6.69	45.82±8.29	41.82±1.83	49.55±4.62	43.11±4.65	47.64±5.24	**50.97**±1.43

For MRSI classification methods, MDL-RS and CCR-Net achieve excellent classification performance on four datasets, such as MDL-RS achieve a higher OA of 95.13%, and yielded over 3% higher than single-modal classification methods [e.g., SVM (75.46%), CDCNN (87.32%), SSRN (93.60%), and DBMA (93.75%)] on the Trento dataset, the reason is that the complementary information is provided by MRSI and thus enhance classification performance. Moreover, the fusion method Co-CNN designed specifically for HSI and LiDAR, and MIFNet designed specifically for HSI and SAR, have achieved favorable classification performance, such as MIFNet achieves the second-highest OA and κ, as well as the highest AA on the Berlin dataset, Co-CNN achieves excellent results that are only surpassed by our proposed method on the Houston 2013 dataset.

By comparing the quantitative classification results of different methods, our proposed model obtains satisfactory classification performance in terms of OA, AA, and κ on the Houston 2013 and Trento datasets, which are far superior to the rest of competitors and achieves the highest OA and κ on both the Augsburg and Berlin datasets. It can be attributed to the fact that IMCL learns feature representations within modalities, and CMCL can capture correspondences between MRSI in the pretraining stage to make the learning more challenging, and further better integration of multimodal complementary information through the proposed HCFM in the fine-tuning stage, thus yielding better classification performance. It can be observed that all methods achieve relatively poor results on the Augsburg and Berlin datasets due to their complex structures. Notably, our proposed model obtains excellent classification performance in terms of OA and κ under limited training samples (only ten per class). While the AA is lower than other methods, which means that our method still has some potential to be explored, and this is the direction of subsequent improvement.

2) Visual Evaluation: The classification maps obtained by different methods are shown in Figs. 5–8. Each of those colors corresponds to a specific land cover category. The DL-based methods have a better visual representation than the traditional method on four datasets. Moreover, the multimodal methods obtain more realistic classification maps due to the integration of complementary information. It can be observed that our proposed model can better distinguish between land cover classes and generate smoother and more realistic classification maps.

D. Ablation Studies

1) Impact of Joint Learning Objective: In the pretraining stage, IMCL and CMCL being learned jointly. IMCL learns intramodal feature representation and CMCL captures the correspondence between multimodal and yields better representation capability. We empirically test the validity by concatenating a fully connected layer classifier behind the feature extractor f_{θ_h} and f_{θ_m}. The OA of all combinations on four datasets is shown in Fig. 9. It can be observed that the classification performance of the learning paradigm of joint IMCL and CMCL is superior to IMCL and CMCL alone. This validates that joint learning objectives can further learn better feature representation.

Moreover, to further analyze the representation capability of the joint learning objective, the feature visualization results on the IMCL, CMCL, and joint IMCL and CMCL learning objective on the Trento dataset is provided. The t-distributed stochastic neighbor embedding (t-SNE) plot of the features is shown in Fig. 10, where (a)–(c) gives the visualization results for the three learning objective of the backbone f_{θ_h}

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:24:13 UTC from IEEE Xplore. Restrictions apply.

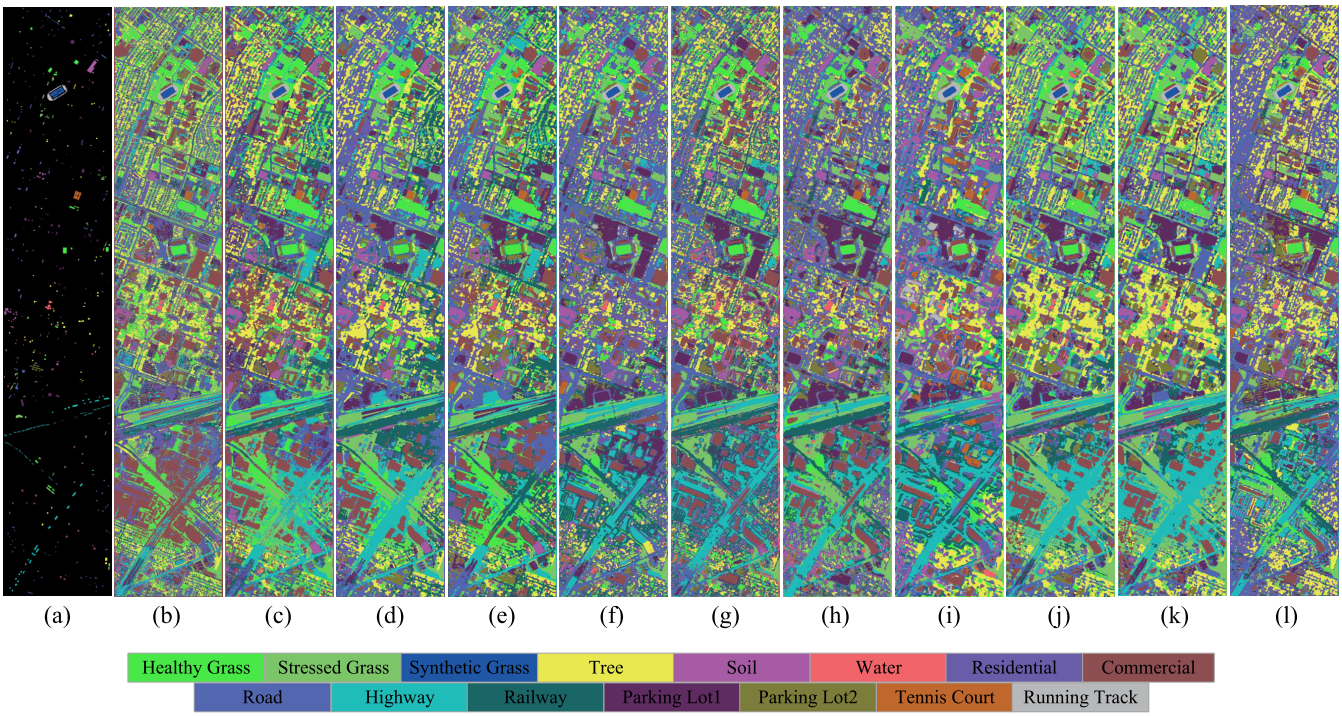


Fig. 5. Classification maps by different methods on the Houston 2013 dataset. (a) False color image. (b) SVM (OA = 74.36%). (c) CDCNN (OA = 78.32%). (d) SSRN (OA = 81.79%). (e) DBMA (OA = 82.16%). (f) Co-CNN (OA = 84.89%). (g) MDL-RS (OA = 84.82%). (h) CCR-Net (OA = 83.25%). (i) MIFNet (OA = 81.62%). (j) SimCLR (OA = 80.80%). (k) SSFR (OA = 84.15%). (l) Proposed (OA = 86.90%).

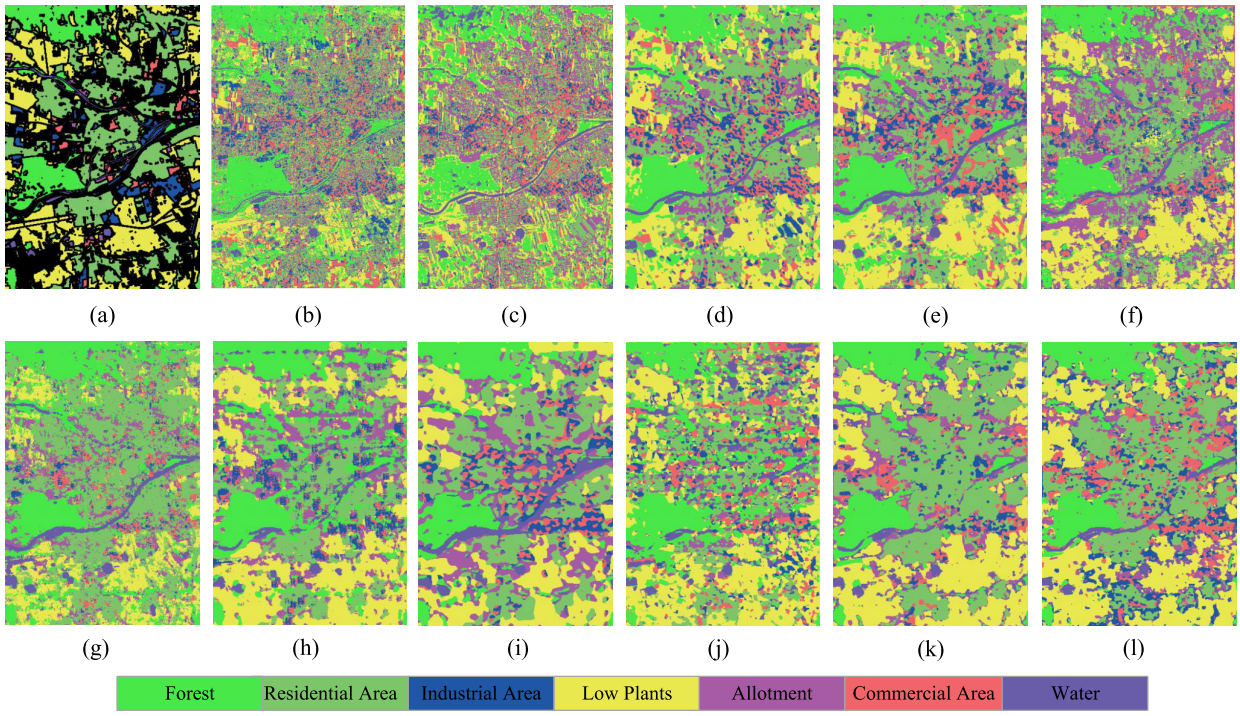


Fig. 6. Classification maps by different methods on the Augsburg dataset. (a) False color image. (b) SVM (OA = 49.53%). (c) CDCNN (OA = 49.59%). (d) SSRN (OA = 74.45%). (e) DBMA (OA = 73.09%). (f) Co-CNN (OA = 74.24%). (g) MDL-RS (OA = 72.53%). (h) CCR-Net (OA = 66.16%). (i) MIFNet (OA = 65.85%). (j) SimCLR (OA = 55.89%). (k) SSFR (OA = 74.36%). (l) Proposed (OA = 79.22%).

of modality A, and the visualization results of the backbone $f_{\theta m}$ of modality B is given by (d)–(f). The joint learning objective can better district classification boundaries compared to individual IMCL and CMCL.

2) Impact of HCFM: In the fine-tuning stage, HCFM is proposed to integrate the complementary information of MRSI further compactly, including LAFM and GAFM. The validity of HCFM is evaluated on four datasets. Fig. 12 graphical illustrates the OA of all combinations. The combination of the LAFM and GAFM outperforms the individual fusion modules. In particular, the combination of the LAFM and GAFM achieved the best OA on the Augsburg dataset, over 6% accuracy gains than individual fusion modules.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:24:13 UTC from IEEE Xplore. Restrictions apply.

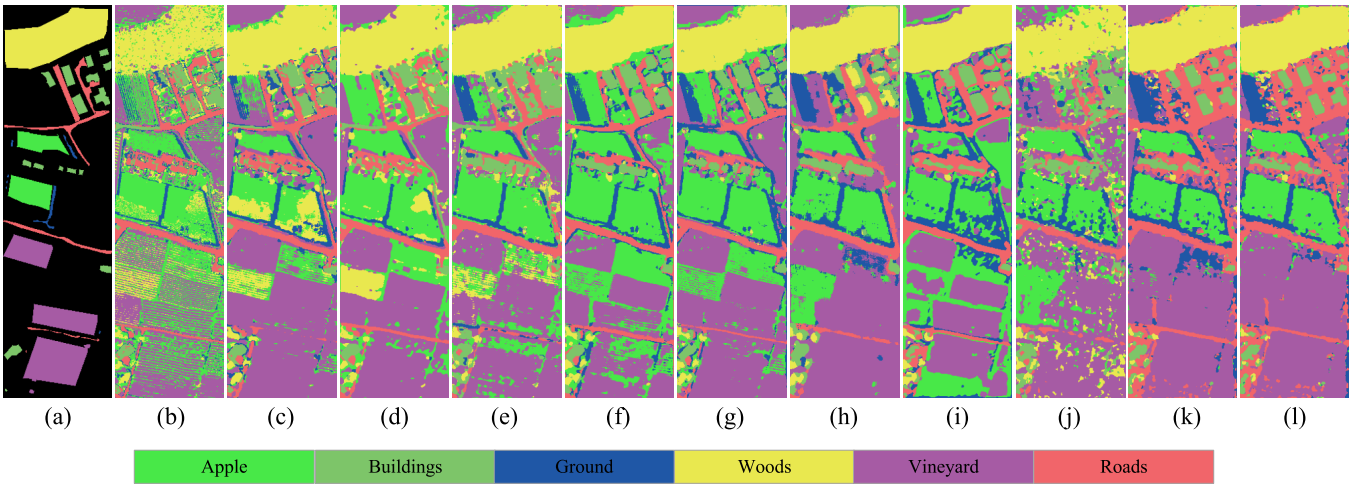


Fig. 7. Classification maps by different methods on the Trento dataset. (a) False color image. (b) SVM (OA = 75.46%). (c) CDCNN (OA = 87.32%). (d) SSRN (OA = 93.60%). (e) DBMA (OA = 93.75%). (f) Co-CNN (OA = 96.68%). (g) MDL-RS (OA = 95.13%). (h) CCR-Net (OA = 97.00%). (i) MIFNet (OA = 92.02%). (j) SimCLR (OA = 79.73%). (k) SSFR (OA = 96.44%). (l) Proposed (OA = 97.62%).

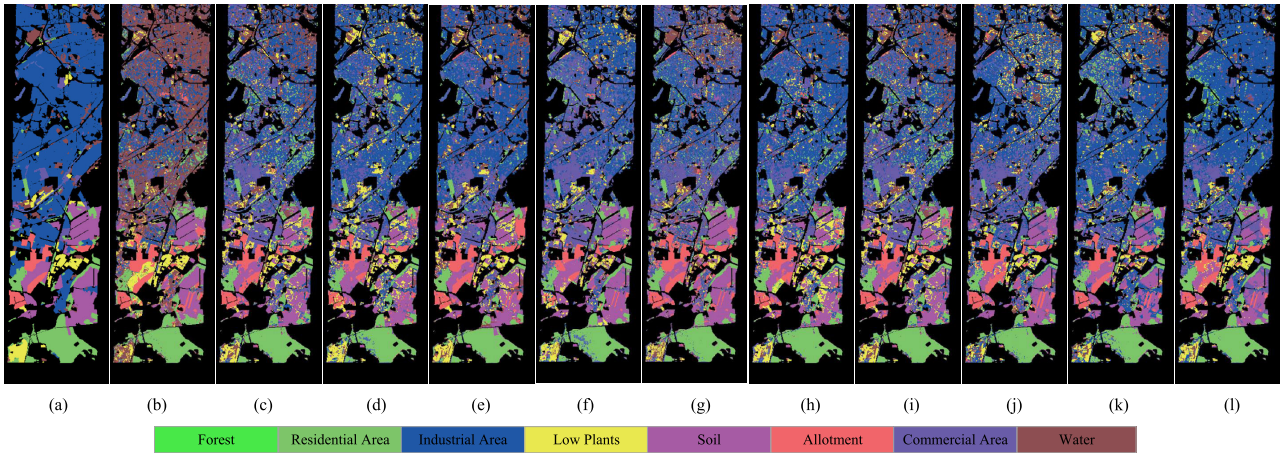


Fig. 8. Classification maps by different methods on the Berlin dataset. (a) False color image. (b) SVM (OA = 47.12%). (c) CDCNN (OA = 48.00%). (d) SSRN (OA = 56.10%). (e) DBMA (OA = 60.15%). (f) Co-CNN (57.16%). (g) MDL-RS (OA = 58.17%). (h) CCR-Net (OA = 54.13%). (i) MIFNet (OA = 61.95%). (j) SimCLR (OA = 56.30%). (k) SSFR (OA = 60.42%). (l) Proposed (OA = 67.35%).

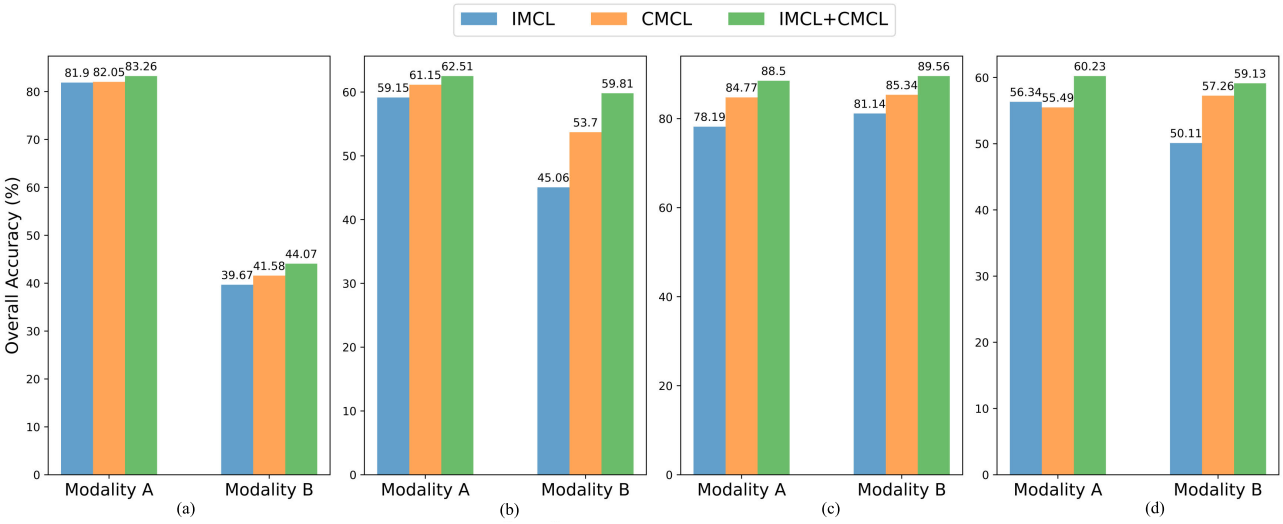


Fig. 9. Ablation studies about the impact of joint learning objective for proposed model on four datasets when compared to IMCL and CMCL. (a) Houston 2013. (b) Augsburg. (c) Trento. (d) Berlin.

3) Impact of Different Number of Labeled Training Samples: To evaluate the impact of the proposed model varying the training samples on the classification performance in the case of limited training samples. In the fine-tuning phase, we finetune the classifier by setting the number of labeled samples between five and 25 per class with intervals of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:24:13 UTC from IEEE Xplore. Restrictions apply.

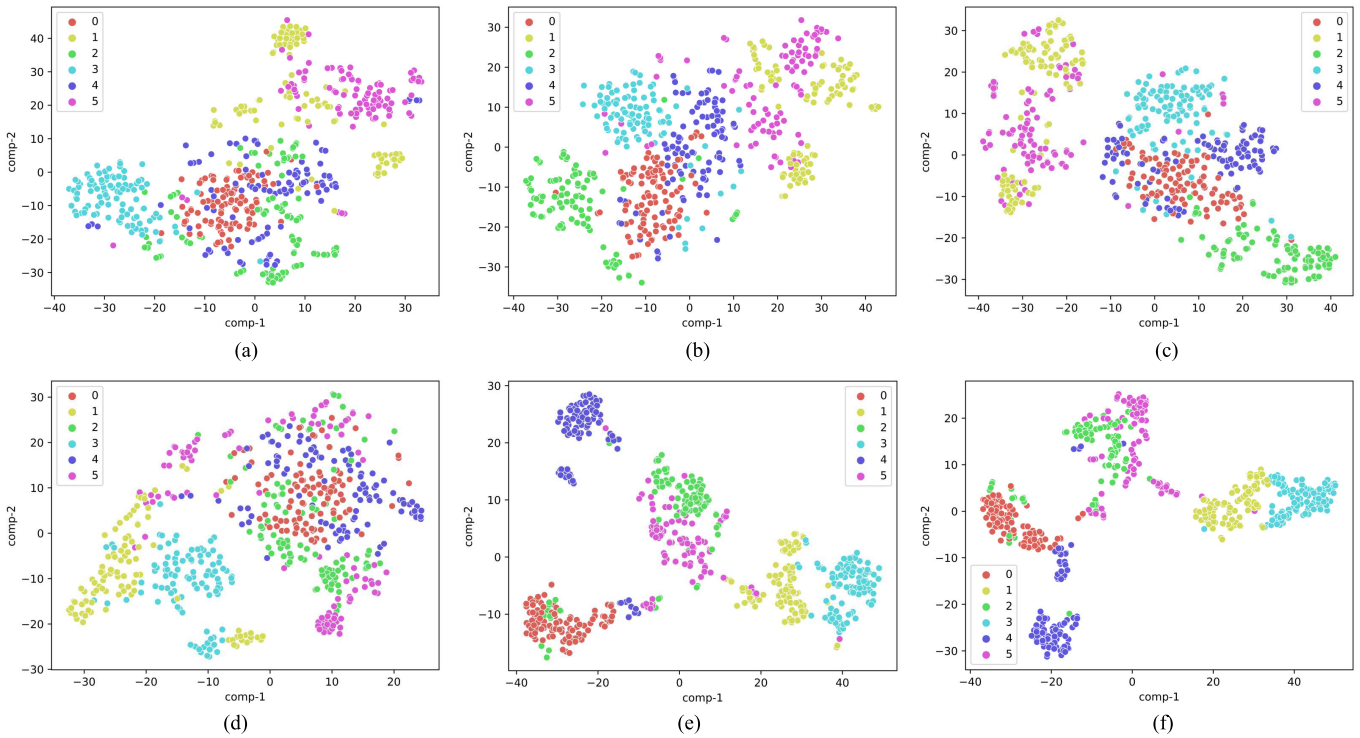


Fig. 10. t-SNE visualization of multimodal features for the backbone f_{θ_h} of modality A and the backbone f_{θ_m} of modality B on the Trento dataset. (a) Modality A: IMCL. (b) Modality A: CMCL. (c) Modality A: IMCL + CMCL. (d) Modality B: IMCL. (e) Modality B: CMCL. (f) Modality B: IMCL + CMCL.

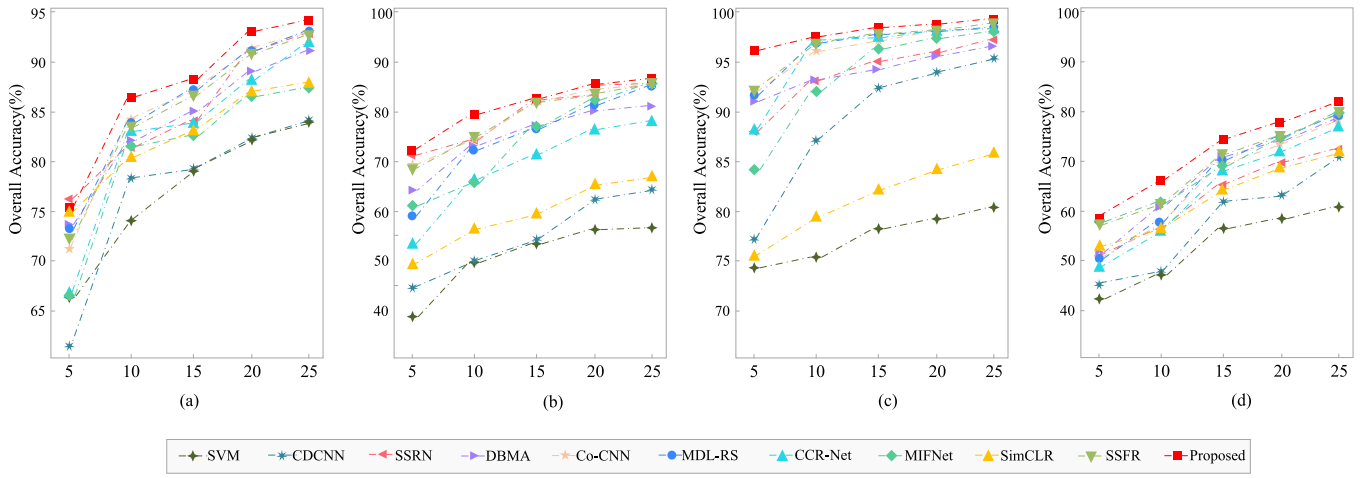


Fig. 11. Ablation studies about the impact of different numbers of labeled training samples for different methods. (a) Houston 2013. (b) Augsburg. (c) Trento. (d) Berlin.

TABLE VII

COMPARISON ON THE NUMBER OF PARAMETERS AND FLOPS FOR DIFFERENT METHODS

Datasets	Complexity	CDCNN [42]	SSRN [43]	DBMA [44]	Co-CNN [19]	MDL-RS [45]	CCR-Net [20]	MIFNet [18]	SimCLR [25]	SSFR [46]	Proposed
Houston 2013	Parameters(M)	0.74	0.27	0.44	0.19	0.19	0.10	0.15	25.62	25.73	0.28
	FLOPs(M)	2.51	169.73	261.41	2.29	2.73	2.41	15.46	32.01	64.24	8.35
Augsburg	Parameters(M)	0.90	0.33	0.54	0.19	0.12	0.11	0.17	25.61	25.73	0.28
	FLOPs(M)	3.11	212.96	329.20	2.39	3.01	2.96	18.02	31.99	64.24	8.41
Trento	Parameters(M)	0.38	0.15	0.20	0.19	0.11	0.08	0.11	25.61	25.73	0.27
	FLOPs(M)	1.15	73.66	129.58	2.29	2.16	1.27	9.81	31.99	172.50	8.34
Berlin	Parameters(M)	1.19	0.43	0.73	0.19	0.13	0.13	0.21	25.61	25.73	0.28
	FLOPs(M)	4.19	289.81	449.73	2.39	3.46	3.87	22.48	32.00	64.24	8.41

five. Fig. 11 illustrates the classification performance of different numbers of labeled training samples with different methods on four datasets. It is visible that the OA of our proposed model steadily improves with the increase of labeled samples and is superior to other methods, which demonstrates the powerful classification performance of our proposed model when the number of training samples is limited.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:24:13 UTC from IEEE Xplore. Restrictions apply.

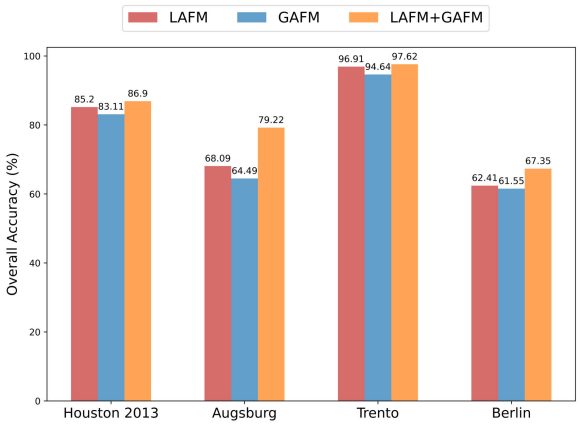


Fig. 12. Ablation studies about the impact of HCFM for proposed model on four datasets.

4) Model Complexity Analysis: We evaluated the complexity of the models using the number of model parameters and floating-point operations (FLOPs). The quantitative comparison results of all methods are listed in Table VII. SimCLR and SSFR use the ResNet-50 deep network model as their feature extractor, which has a high number of parameters and FLOPs. SSRN and DBMA have higher FLOPs due to the predominant use of 3-D convolutions in their networks. In contrast, our method has a relatively low number of parameters and FLOPs because our feature extractor only uses three successive 2-D convolutions.

V. Conclusion

This article proposes an effective self-supervised CMCL method for MRSI classification under limited samples. The joint IMCL and CMCL to better mine multimodal feature representations. Moreover, a simple and effective HCFM is proposed in the fine-tuning stage, which can better fuse heterogeneous features. The classification performance of the proposed model is validated on four datasets, and ablation experiments validate the effectiveness of joint IMCL and CMCL objective and the efficacy of the HCFM. However, our method requires that the datasets used in the pretraining and fine-tuning phases obey the same distribution. Also, since our method requires a longer time for pretraining, it may not be suitable for some real-time applications. In the future, we need to consider compressing the pretraining time without compromising performance.

References

[1] B. Zhang, X. Sun, L. Gao, and L. Yang, "Endmember extraction of hyperspectral remote sensing images based on the ant colony optimization (ACO) algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 7, pp. 2635–2646, Jul. 2011.

[2] A. Ibrahim et al., "Atmospheric correction for hyperspectral ocean color retrieval with application to the hyperspectral imager for the coastal ocean (HICO)," *Remote Sens. Environ.*, vol. 204, pp. 60–75, Jan. 2018.

[3] X. Yao, X. Feng, J. Han, G. Cheng, and L. Guo, "Automatic weakly supervised object detection from high spatial resolution remote sensing images via dynamic curriculum learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 675–685, Jan. 2021.

[4] D. Hong, J. Yao, D. Meng, Z. Xu, and J. Chanussot, "Multimodal GANs: Toward crossmodal hyperspectral–multispectral image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5103–5113, Jun. 2021.

[5] Z. Li, L. Huang, and J. He, "A multiscale deep middle-level feature fusion network for hyperspectral classification," *Remote Sens.*, vol. 11, no. 6, p. 695, Mar. 2019.

[6] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "CoSpace: Common subspace learning from hyperspectral-multispectral correspondences," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4349–4359, Jul. 2019.

[7] L.-Z. Huo et al., "Supervised spatial classification of multispectral LiDAR data in urban areas," *PLoS ONE*, vol. 13, no. 10, Oct. 2018, Art. no. e0206185.

[8] U. Heiden, W. Heldens, S. Roessner, K. Segl, T. Esch, and A. Mueller, "Urban structure type characterization using hyperspectral remote sensing and height information," *Landscape Urban Planning*, vol. 105, no. 4, pp. 361–375, Apr. 2012.

[9] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag. Replaces Newsletter*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[10] L. Gómez-Chova, D. Tuia, G. Moser, and G. Camps-Valls, "Multimodal classification of remote sensing images: A review and future directions," *Proc. IEEE*, vol. 103, no. 9, pp. 1560–1584, Sep. 2015.

[11] M. Zhang, W. Li, Q. Du, L. Gao, and B. Zhang, "Feature extraction for classification of hyperspectral and LiDAR data using patch-to-patch CNN," *IEEE Trans. Cybern.*, vol. 50, no. 1, pp. 100–111, Jan. 2020.

[12] H. Su, Y. Yu, Q. Du, and P. Du, "Ensemble learning for hyperspectral image classification using tangent collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3778–3790, Jun. 2020.

[13] B. Huang, Y. Li, X. Han, Y. Cui, W. Li, and R. Li, "Cloud removal from optical satellite imagery with SAR imagery using sparse representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 5, pp. 1046–1050, May 2015.

[14] M. Pedergnana, P. R. Marpu, M. D. Mura, J. A. Benediktsson, and L. Bruzzone, "Classification of remote sensing optical and LiDAR data using extended attribute profiles," *IEEE J. Sel. Topics Signal Process.*, vol. 6, no. 7, pp. 856–865, Nov. 2012.

[15] W. Liao, A. Pižurica, R. Bellens, S. Gautama, and W. Philips, "Generalized graph-based fusion of hyperspectral and LiDAR data using morphological features," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 3, pp. 552–556, Mar. 2015.

[16] B. Rasti, P. Ghamisi, J. Plaza, and A. Plaza, "Fusion of hyperspectral and LiDAR data using sparse and low-rank component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6354–6365, Nov. 2017.

[17] B. Rasti, P. Ghamisi, and R. Gloaguen, "Hyperspectral and LiDAR fusion using extinction profiles and total variation component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3997–4007, Jul. 2017.

[18] J. Wang, W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Hyperspectral and SAR image classification via multiscale interactive fusion network," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, May 11, 2022, doi: 10.1109/TNNLS.2022.3171572.

[19] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[20] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.

[21] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[22] S. K. Roy, A. Deria, D. Hong, M. Ahmad, A. Plaza, and J. Chanussot, "Hyperspectral and LiDAR data classification using joint CNNs and morphological feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530416.

[23] W. Dong, T. Zhang, J. Qu, S. Xiao, T. Zhang, and Y. Li, "Multibranch feature fusion network with Self- and cross-guided attention for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530612.

[24] Y. Zhang, H. L. Yang, S. Prasad, E. Pasolli, J. Jung, and M. Crawford, "Ensemble multiple kernel active learning for classification of multisource remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 2, pp. 845–858, Feb. 2015.

[25] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2020, pp. 1597–1607.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:24:13 UTC from IEEE Xplore. Restrictions apply.

[26] J.-B. Grill et al., "Bootstrap your own latent-a new approach to self-supervised learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 21271–21284.
[27] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9726–9735.
[28] B. Liu, A. Yu, X. Yu, R. Wang, K. Gao, and W. Guo, "Deep multiview learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7758–7772, Sep. 2021.
[29] M. Zhu, J. Fan, Q. Yang, and T. Chen, "SC-EADNet: A self-supervised contrastive efficient asymmetric dilated network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519517.
[30] J. Feng, N. Zhao, R. Shang, X. Zhang, and L. Jiao, "Self-supervised divide-and-conquer generative adversarial network for classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536517.
[31] L. Song, Z. Feng, S. Yang, X. Zhang, and L. Jiao, "Self-supervised assisted semi-supervised residual network for hyperspectral image classification," *Remote Sens.*, vol. 14, no. 13, p. 2997, Jun. 2022.
[32] Y. Chen, C. Li, P. Ghamisi, X. Jia, and Y. Gu, "Deep fusion of remote sensing data for accurate classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1253–1257, Aug. 2017.
[33] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.
[34] H. Xu, W. He, L. Zhang, and H. Zhang, "Unsupervised spectral–spatial semantic feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526714.
[35] J. Xia, N. Yokoya, and A. Iwasaki, "Fusion of hyperspectral and LiDAR data with a novel ensemble classifier," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 6, pp. 957–961, Jun. 2018.
[36] J. Xia, W. Liao, and P. Du, "Hyperspectral and LiDAR classification with semisupervised graph fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 4, pp. 666–670, Apr. 2020.
[37] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.
[38] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[39] K. Ding, T. Lu, W. Fu, S. Li, and F. Ma, "Global–local transformer network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541213.
[40] W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Asymmetric feature fusion network for hyperspectral and SAR image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Feb. 18, 2022, doi: 10.1109/TNNLS.2022.3149394.
[41] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.
[42] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.
[43] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.
[44] W. Ma, Q. Yang, Y. Wu, W. Zhao, and X. Zhang, "Double-branch multi-attention mechanism network for hyperspectral image classification," *Remote Sens.*, vol. 11, no. 11, p. 1307, Jun. 2019.
[45] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.
[46] Z. Xue, B. Liu, A. Yu, X. Yu, P. Zhang, and X. Tan, "Self-supervised feature representation and few-shot land cover classification of multimodal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541618.
[47] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[48] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.



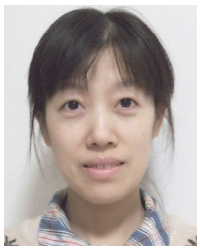
Zhixi Feng (Member, IEEE) received the B.S. degree in automation from the Lanzhou University of Technology, Lanzhou, China, in 2012, and the Ph.D. degree in intelligent information processing from Xidian University, Xi'an, China, in 2018.

He is currently an Associate Professor of artificial intelligence with Xidian University. His research interests include machine learning and remote sensing image interpretation.



Liangliang Song received the B.S. degree from Shanxi University, Taiyuan, China, in 2021. He is currently pursuing the master's degree with the School of Artificial Intelligence, Xidian University, Xi'an, China.

His research interests include image fusion and deep learning.



Shuyuan Yang (Senior Member, IEEE) received the B.S. degree in electrical engineering, and the M.S. and Ph.D. degrees in circuit and system from Xidian University, Xi'an, China, in 2000, 2003, and 2005, respectively.

She is currently a Professor with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education of China, Xidian University. Her main research interests include intelligent signal processing, machine learning, and image processing



Xinyu Zhang received the B.S. degree from Central China Normal University, Wuhan, China, in 2021. He is currently pursuing the master's degree with the School of Artificial Intelligence, Xidian University, Xi'an, China.

His research interests include image processing and machine learning.



Licheng Jiao (Fellow, IEEE) received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

Since 1992, he has been a Professor with the School of Electronic Engineering, Xidian University, Xi'an, where he is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China. He is in charge of about 40 important scientific research projects. He has authored/coauthored more than 20 monographs and 100 papers in international journals and conferences. His research interests include image processing, natural computation, machine learning, and intelligent information processing.

Dr. Jiao is a Foreign Member of the Academia Europaea and the Russian Academy of Natural Sciences; a fellow of the Institution of Engineering and Technology (IET), the Chinese Association for Artificial Intelligence (CAAI), the Chinese Institute of Electronics (CIE), the China Computer Federation (CCF), and the Chinese Association of Automation (CAA); a Councilor of CIE; a Committee Member of the Chinese Committee of Neural Networks; an Expert of the Academic Degrees Committee of the State Council; the Chairperson of the Awards and Recognition Committee; and the Vice Board Chairperson of the Chinese Association of Artificial Intelligence.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:24:13 UTC from IEEE Xplore. Restrictions apply.